

Abhinav Raj

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Education

University of Illinois Urbana-Champaign (UIUC)

Bachelor of Science in Computer Science & Linguistics

Champaign, Illinois

Expected Graduation: May 2029

- GPA: 4.0/4.0 (Dean's List)
- Concentrations planned: Computer Science, Linguistics, Statistics
- Relevant coursework: Discrete Structures; Calculus III; Statistics and Probability I

Technical Skills

- **Languages/Tools:** Rust, C++, Java, Python (NumPy, Pandas, Matplotlib), Git, HTML/CSS

Experience

Financial Engineering Club

Member

Champaign, Illinois

Feb 2026 – Present

- Built a Python research pipeline comparing Buy & Hold, 252-day Time-Series Momentum, Directional Change, and Multi-Threshold DC on SPY/QQQ/TLT/GLD daily ETF data from 2010–2026.
- Implemented one-day-lagged backtests with normalized exposures, turnover-based transaction costs, and 0/5/10 bps robustness checks; tracked return, Sharpe, volatility, max drawdown, turnover, and trades.
- Showed Time-Series Momentum was the strongest active strategy, earning 76.62% total return with 0.41 Sharpe and 3.73% turnover, while higher-turnover DC variants underperformed after costs.

SIGNLL — Speech and Interdisciplinary Group for Natural Language Learning

Research Member

Champaign, Illinois

Aug 2025 – Present

- Compared small language models (SLMs) and large language models (LLMs) for specialized tasks, focusing on quality, latency, and cost trade-offs.
- Designed evaluation prompts and test cases for domain-specific functions; tracked performance with task-level metrics and error analysis.
- Ran experiments across model variants and inference settings; summarized results into clear takeaways for downstream use.
- Communicated findings with the team and iterated on experimental design based on observed failure modes.

Solar Car Team Illinois

Strategy Team Member

Champaign, Illinois

Aug 2025 – Dec 2025

- Used Python and SQL to analyze 240,000+ lines of telemetry related to battery voltage, power, and current.
- Parsed JSON logs and visualized trends with Pandas/Polars/NumPy and Matplotlib to support data-quality improvements.

Projects

SilentSync

2023 – 2025

- Launched an app to support deaf employees and improve workplace communication.
- Conducted 100+ user interviews; iterated on product based on feedback from deaf users and retail partners.
- Secured partnerships with local businesses, increasing usage by 40% and reaching 150+ downloads in the first month.

NYC Traffic Fatality Prediction

June 2024

- Built ML models on 74,881 NYC traffic accident records and achieved 99.82% predictive accuracy for fatalities.
- Applied feature engineering and imbalance handling (one-hot encoding, SMOTE) and documented error analysis.
- Evaluated classification quality with precision/recall/F1 and a confusion matrix; reviewed error slices by borough, time-of-day, and weather conditions.

Publications

- **Deaf Accessibility to Public Spaces** (Feb 2024) — DOI: 10.14445/23942703/IJHSS-V11I5P109; presented at the Innosphere Research Conference.
- Synthesized 30+ sources; proposed actionable interventions (signage, staff training, assistive tech) for inclusive public communication.